

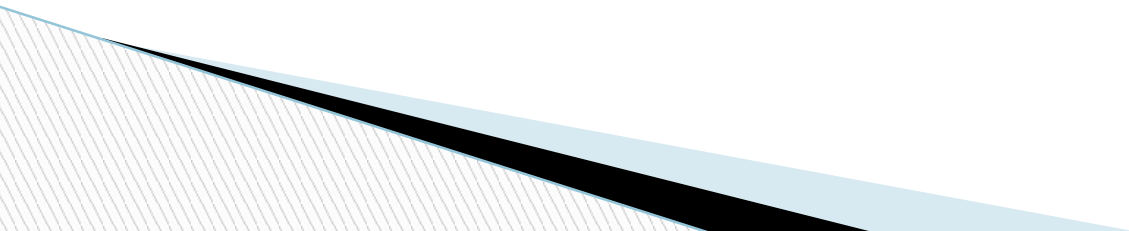


Overview of Supervised & Un-Supervised Learning

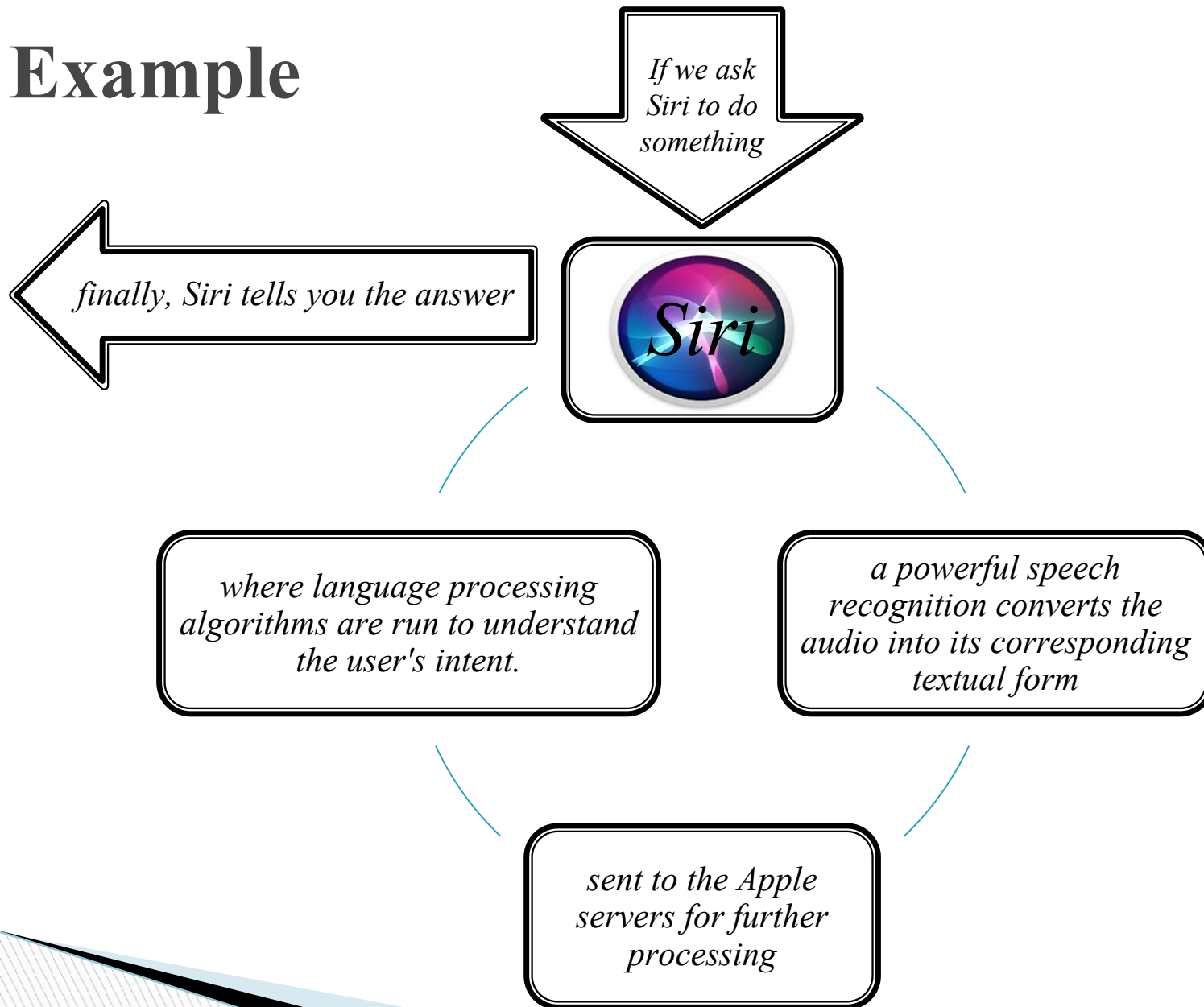
Dr.G.Padmavathi
Assistant Professor
CRRAO AIMSCS

Introduction to Machine learning

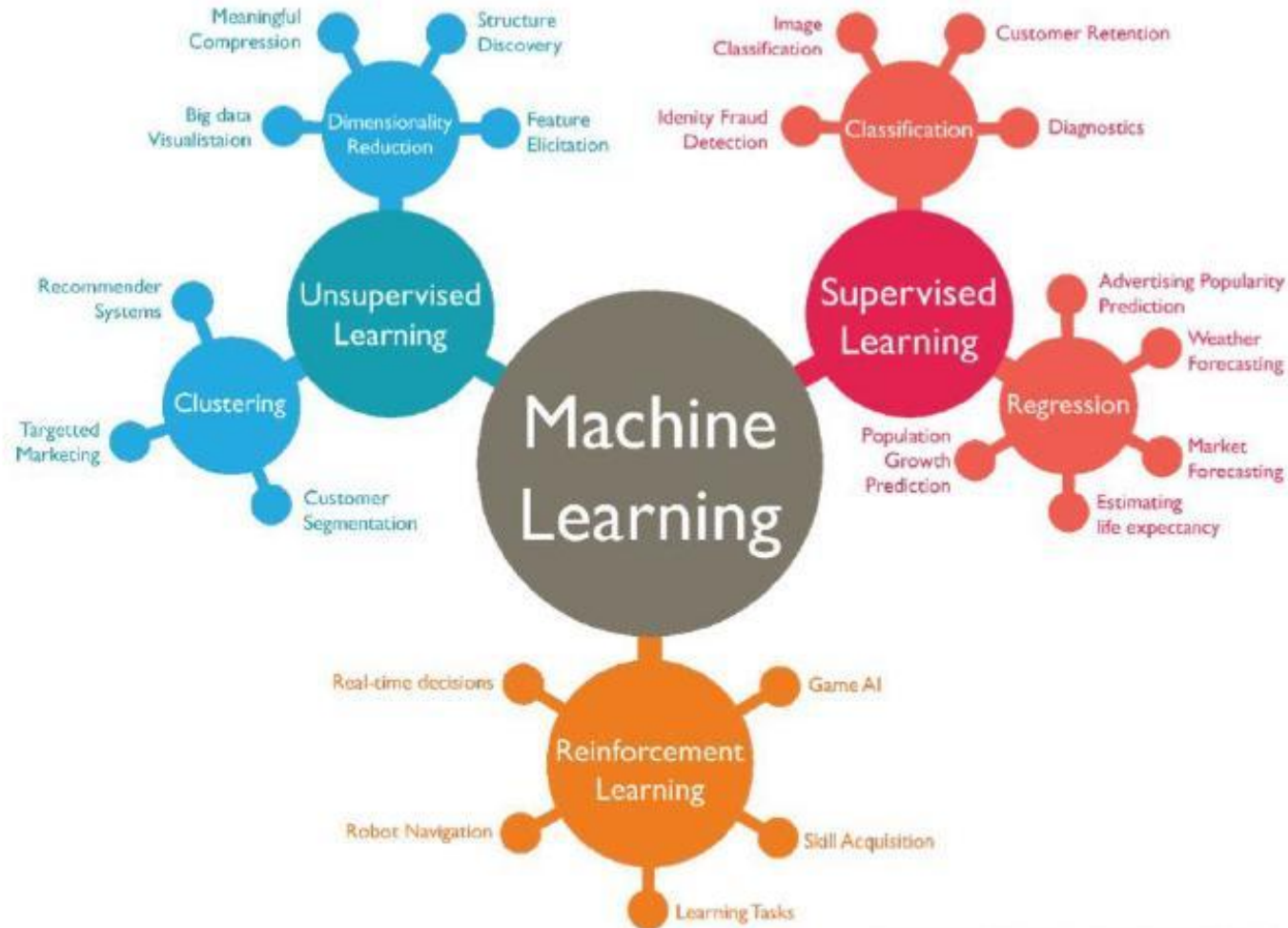
- Machine learning algorithms are trained with training data.
- When new data comes in, they can make predictions and decisions accurately based on past data.



Example



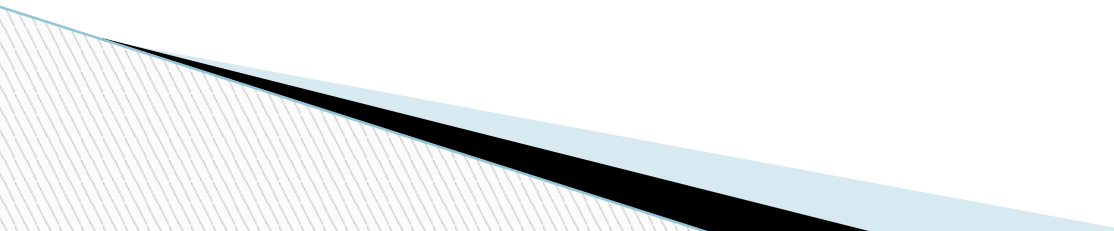
TYPES OF ML/DL



source nvidia via @mikequindazzi

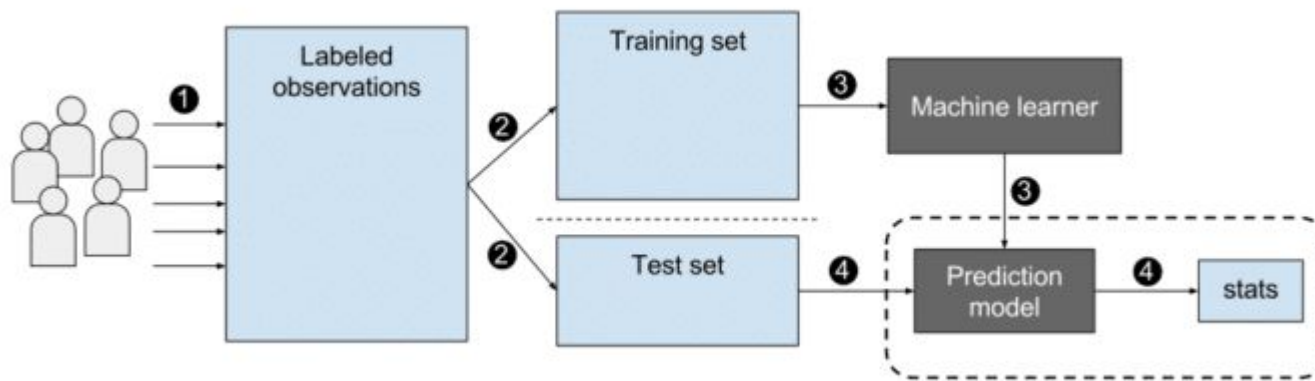
<https://livedeestream.dev/post/2020-06-02-7-steps-of-machine-learning/>

Types of machine learning

- Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learning
- 

Supervised Learning

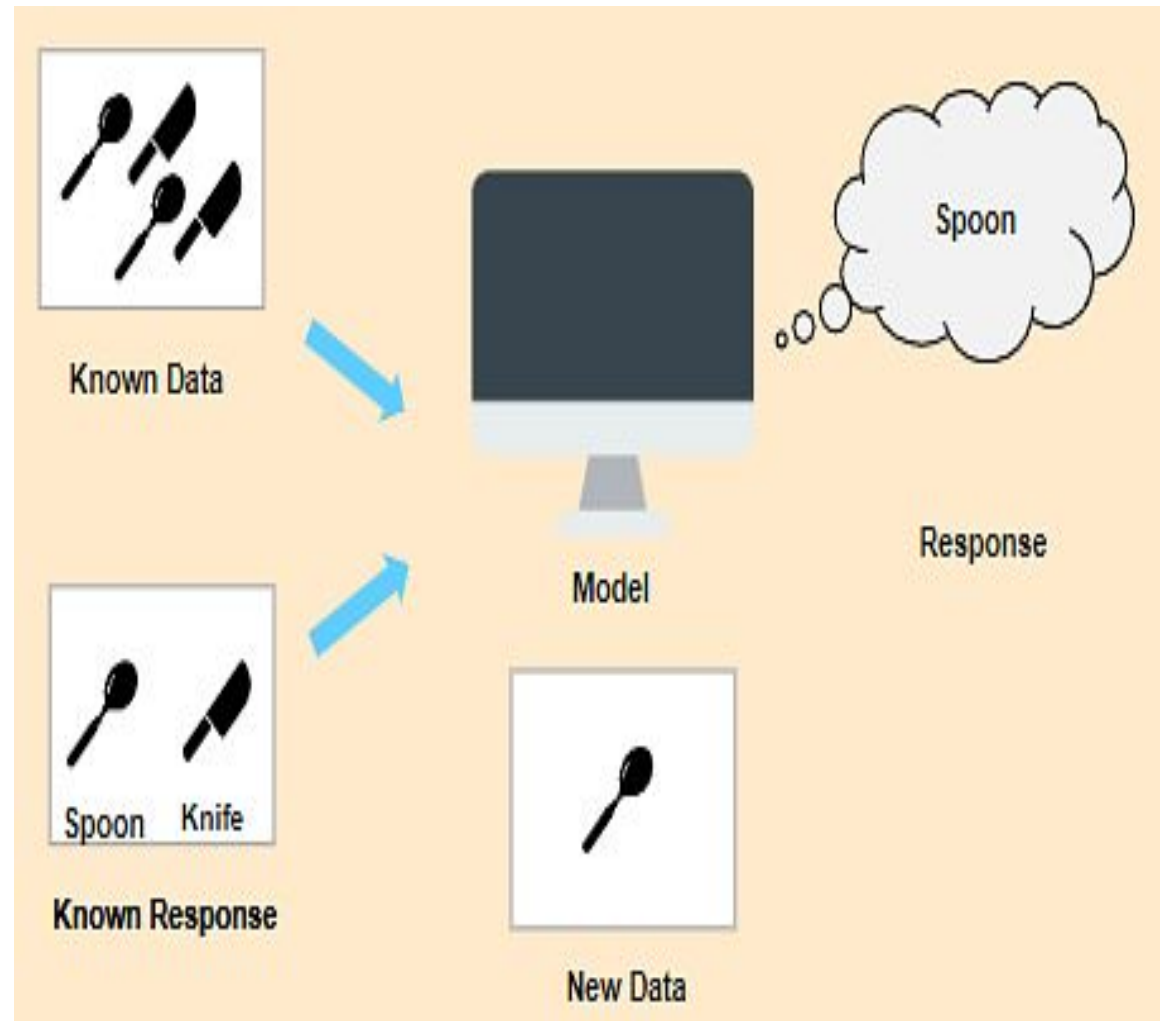
- In Supervised Learning, the machine learns under supervision.
- It contains a model that is able to predict with the help of a labeled dataset. A labeled dataset is one where you already know the target answer.



Example of Supervised Learning

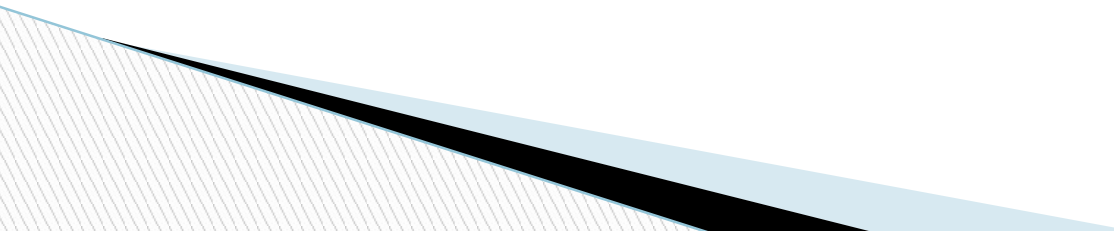
Known data is fed to the machine, which analyzes and learns the association of these images based on its features such as shape, size, sharpness, etc.

When a new image is fed to the machine without any label, the machine is able to predict accurately that it is a spoon with the help of the past data.

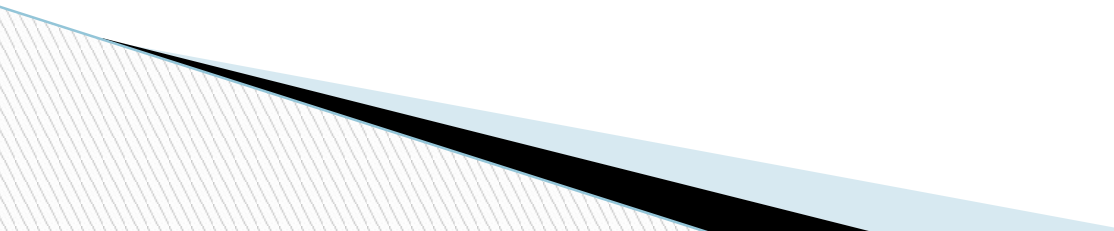


Classification of Data

Types of ML Classification Algorithms:

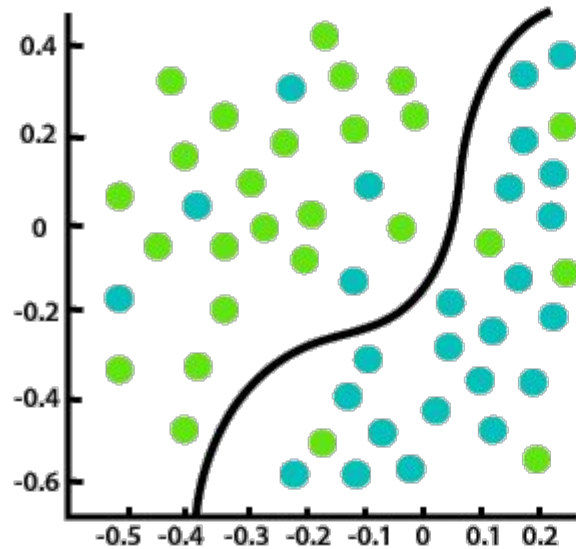
- ❑ Logistic Regression
 - ❑ K-Nearest Neighbours
 - ❑ Support Vector Machines
 - ❑ Kernel SVM
 - ❑ Naïve Bayes
 - ❑ Decision Tree Classification
 - ❑ Random Forest Classification
- 

Types of Regression Algorithm

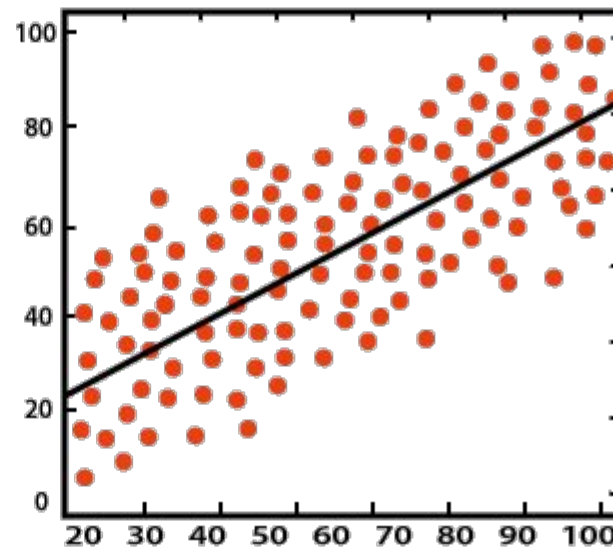
- Simple Linear Regression
 - Multiple Linear Regression
 - Polynomial Regression
 - Support Vector Regression
 - Decision Tree Regression
 - Random Forest Regression
- 

Regression Algorithm	Classification Algorithm
In Regression, the output variable must be of continuous nature or real value.	In Classification, the output variable must be a discrete value.
The task of the regression algorithm is to map the input value (x) with the continuous output variable(y).	The task of the classification algorithm is to map the input value(x) with the discrete output variable(y).
Regression Algorithms are used with continuous data.	Classification Algorithms are used with discrete data.
In Regression, we try to find the best fit line, which can predict the output more accurately.	In Classification, we try to find the decision boundary, which can divide the dataset into different classes.
Regression algorithms can be used to solve the regression problems such as Weather Prediction, House price prediction, etc.	Classification Algorithms can be used to solve classification problems such as Identification of spam emails, Speech Recognition, Identification of cancer cells, etc.
The regression Algorithm can be further divided into Linear and Non-linear Regression.	The Classification algorithms can be divided into Binary Classifier and Multi-class Classifier.

Regression and Classification

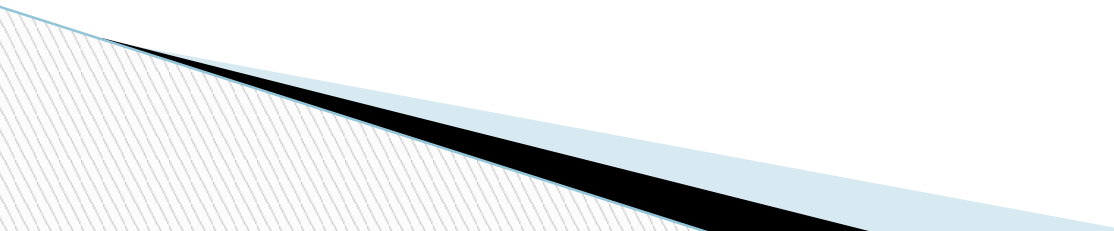


Classification



Regression

Classifier comparison

- A comparison of a several classifiers in scikit-learn on synthetic datasets.
 - The point of this example is to illustrate the nature of decision boundaries of different classifiers.
 - The plots show training points in solid colors and testing points semi-transparent. The lower right shows the classification accuracy on the test set.
- 

The plots show training points in solid colors and testing points semi-transparent. The lower right shows the classification accuracy on the test set.

Input data

Nearest Neighbors

Linear SVM

RFB SVM

Gaussian Process

Decision Tree

Random Forest

Neural Net

AdaBoost

Naive Bayes

QDA

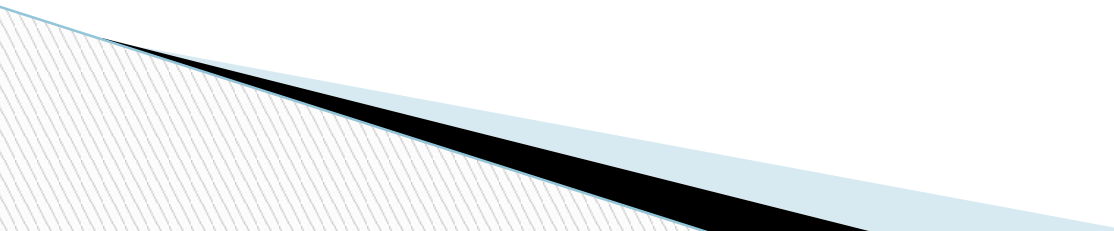


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Unsupervised Learning

- Unsupervised machine learning finds all kind of unknown patterns in data.
- Unsupervised methods help you to find features which can be useful for categorization.
- It is taken place in real time, so all the input data to be analyzed and labeled in the presence of learners.
- It is easier to get unlabeled data from a computer than labeled data, which needs manual intervention.

where you are not taught but you learn from the data (based on features/patterns/associations) - comes under Unsupervised Learning.



Clustering-Unsupervised Learning



sample



Cluster/group

There are different types of clustering we can utilize:

Exclusive (partitioning)

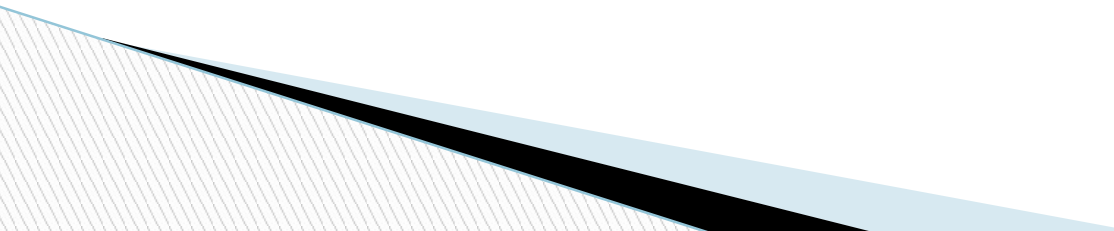
- In this clustering method, Data are grouped in such a way that one data can belong to one cluster only.

Example: K-means

Agglomerative

- In this clustering technique, every data is a cluster. The iterative unions between the two nearest clusters reduce the number of clusters.

Example: Hierarchical clustering



Overlapping

- In this technique, fuzzy sets is used to cluster data. Each point may belong to two or more clusters with separate degrees of membership.
- Here, data will be associated with an appropriate membership value.
Example: Fuzzy C-Means

Probabilistic

- This technique uses probability distribution to create the clusters

Example: Following keywords can be clustered into two categories “shoe” and “glove” or “man” and “women.”

- “man’s shoe.”
 - “women’s shoe.”
 - “women’s glove.”
 - “man’s glove.”
- 

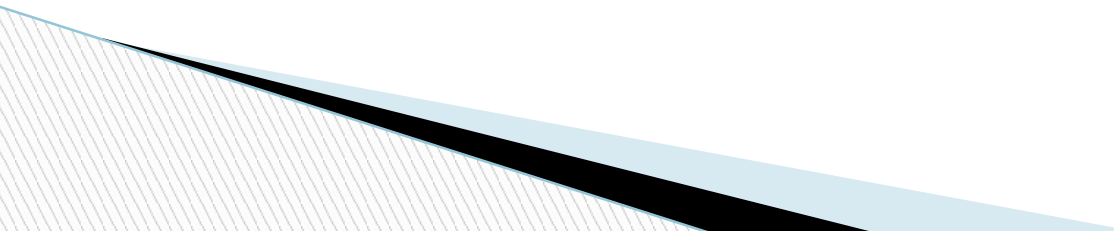
Clustering Types

- Hierarchical clustering
 - K-means clustering
 - K-NN (k nearest neighbors)
 - Principal Component Analysis
 - Singular Value Decomposition
 - Independent Component Analysis
-
- [A Comprehensive Survey of Clustering Algorithms](#) Xu, D. & Tian, Y. Ann. Data. Sci. (2015) 2: 165.
 - <https://developers.google.com/machine-learning/clustering/overview>

Association-Unsupervised Learning

- Association rules allow us to establish associations amongst data objects inside large databases. This unsupervised technique is about discovering interesting relationships between variables in large databases.

Some examples

- people that buy a new home most likely to buy new furniture.
 - Groups of shopper based on their browsing and purchasing histories
 - Movie group by the rating given by movies viewers
 - A subgroup of cancer patients grouped by their gene expression measurements
- 

Applications of Unsupervised Machine Learning

- ❑ Clustering automatically split the dataset into groups base on their similarities
- ❑ Anomaly detection can discover unusual data points in our dataset. It is useful for finding fraudulent transactions.
- ❑ Association mining identifies sets of items which often occur together in our dataset
- ❑ Latent variable models are widely used for data preprocessing. Like reducing the number of features in a dataset or decomposing the dataset into multiple components
- ❑ <https://www.kaggle.com/farhanmd29/unsupervised-learning>

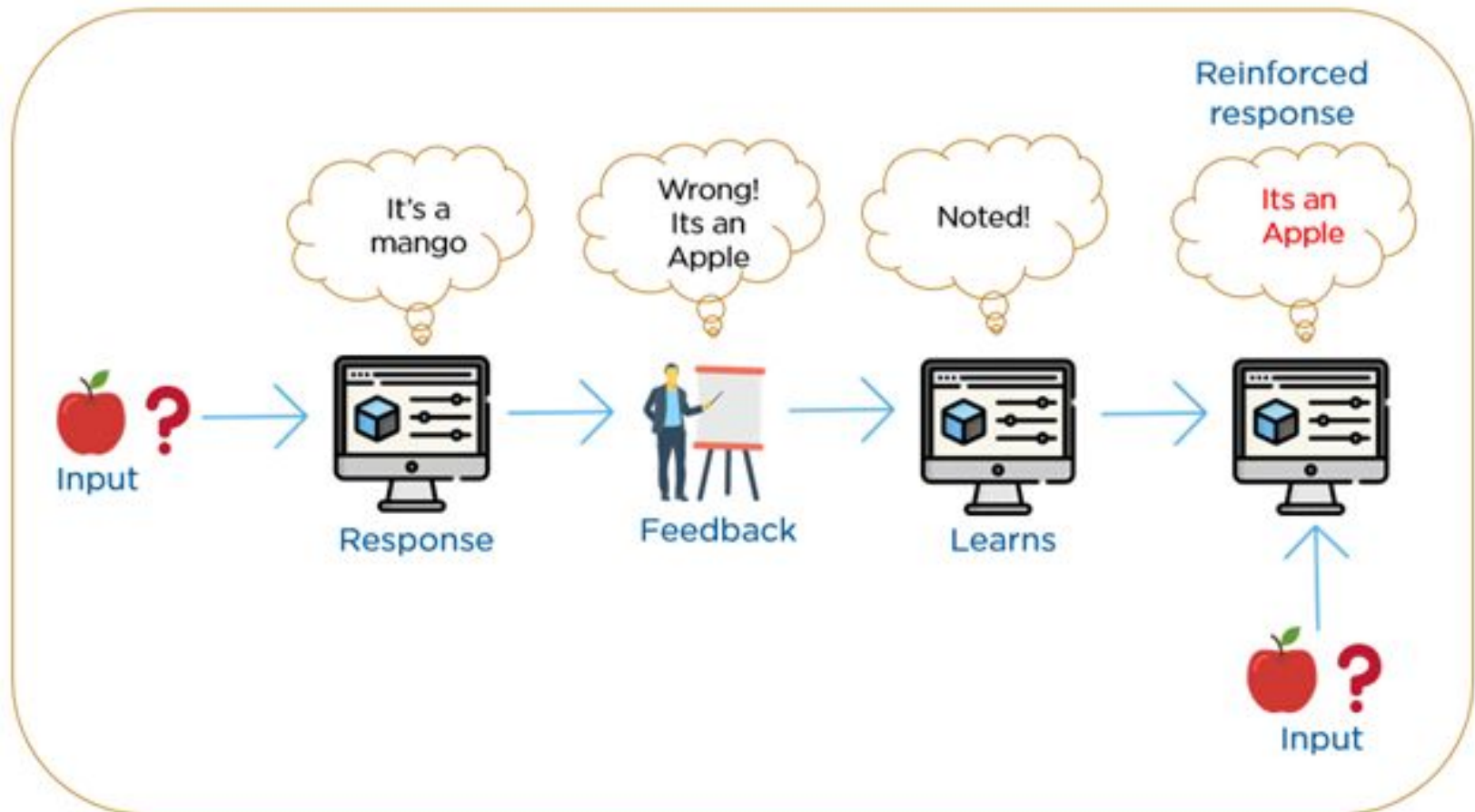
Supervised vs. Unsupervised Machine Learning

Parameters	Supervised machine learning technique	Unsupervised machine learning technique
Input Data	Algorithms are trained using labeled data.	Algorithms are used against data which is not labelled
Computational Complexity	Supervised learning is a simpler method.	Unsupervised learning is computationally complex
Accuracy	Highly accurate and trustworthy method.	Less accurate and trustworthy method.

Reinforcement learning

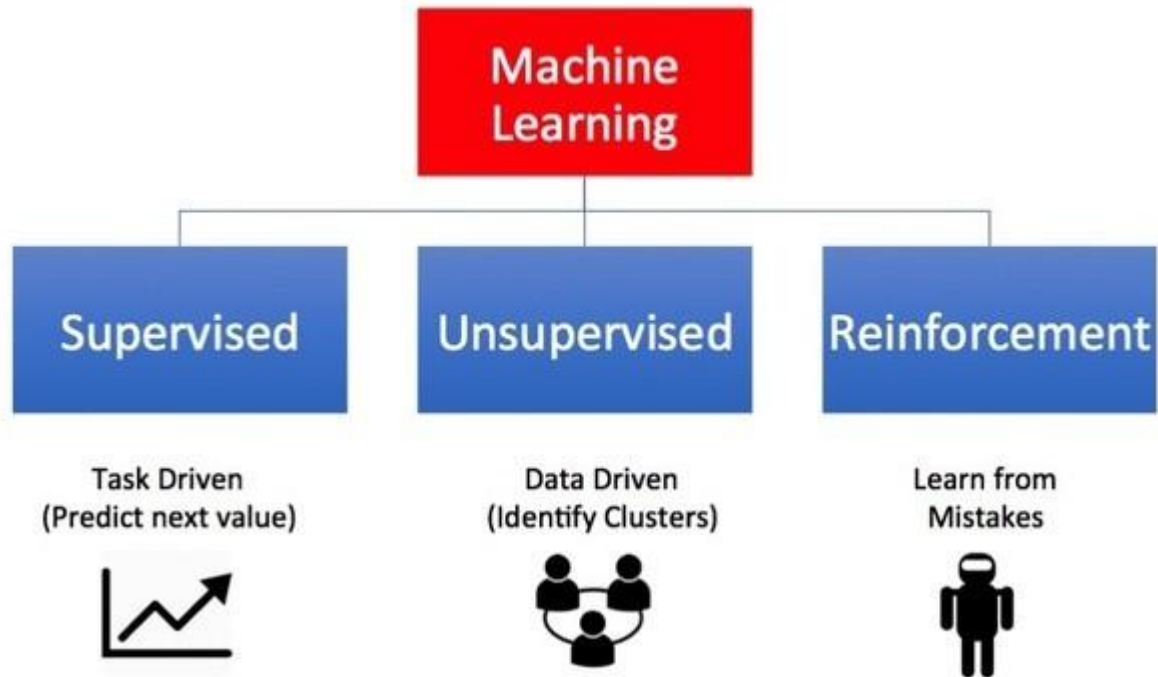
- Reinforcement Learning(RL) is a type of machine learning technique that enables an agent to learn in an interactive environment by trial and error using feedback from its own actions and experiences.

Reinforcement learning



<https://towardsdatascience.com/an-introduction-to-reinforcement-learning-4e7825c60bbe>

Types of Machine Learning



<https://towardsdatascience.com/reinforcement-learning-101-e24b50e1d292>

Thank you

Refere

1. <https://blogs.nvidia.com/blog/2018/08/02/supervised-unsupervised-learning/#:~:text=In%20a%20supervised%20learning%20model,and%20patterns%20on%20its%20own.>
2. <https://www.javatpoint.com/regression-vs-classification-in-machine-learning>
3. <https://towardsdatascience.com/introduction-to-regression-analysis-9151d8ac14b3#:~:text=Regression%20and%20classification%20are%20both,their%20models%20and%20make%20predictions.&text=While%20in%20regression%20C%20the%20output,output%20is%20categorical%20or%20discrete.>
4. <https://www.guru99.com/unsupervised-machine-learning.html>